

FinGear AI: An Explainable Financial Digital Twin for Financial Health Scoring, Anomaly Detection and Forecasting

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Abstract— FinGear AI is an explainable personal financial intelligence application designed as a Financial Digital Twin for customized financial wellness, forecasting, and decision support. Users can securely save financial profiles, record income and expenses, attach PDF bills, track transactions, and recover recently deleted records. Users are classified into income tiers and transactions into Needs, Wants, and Savings. Instead of a fixed 50:30:20 rule, FinGear AI applies adaptive budgeting targets based on income level. It assesses spending distribution, cash flow, emergency reserves, debt load, saving behavior, investment consistency, and payment discipline to produce an interpretable financial health score with component-level reasons, risk alerts, and actionable recommendations. Abnormal expenditure transactions are detected, and users can confirm normal records or exclude irregular ones from the model's learning. Personalized expense and cash-flow forecasts begin with rule-based estimates for new users and transition to user-specific machine-learning and ARIMA models as transaction history grows. The Financial Digital Twin unifies profile data, transaction behavior, health indicators, anomaly signals, and forecasts to support budgeting, goal setting, AI Copilot guidance, what-if simulations, and investment decision support. Built with React, FastAPI, PostgreSQL, and machine-learning services, it offers a secure, transparent, and scalable platform for better personal financial decision making.

Keywords—Financial digital twin, explainable AI, financial health score, anomaly detection, ARIMA, random forest, personal finance management, income-tier budgeting.

I. INTRODUCTION

Financial wellness depends on a person's ability to meet current commitments, absorb financial shocks, manage borrowing and make progress toward future goals. These outcomes are not visible from income alone; a high-income individual may face unstable cash flow or inadequate reserves, while a lower-income individual may be resilient through disciplined behaviour. A useful personal-finance system must therefore analyse behaviour in context rather than impose a single generic rule.

Existing personal-finance applications largely record income, expenses and balances, offering fragmented and reactive insight. Users may see how much they spent in a category without understanding whether the pattern is sustainable, or how it affects their emergency fund, debt position or long-term goals. Generic rules such as the fixed 50:30:20 allocation cannot account for differences in income, dependents, debt obligations and financial priorities.

FinGear AI addresses this gap through an explainable Financial Digital Twin. This paper reports the Phase I assessment layer of the system, which is limited in scope to three tightly connected modules: (i) an adaptive, explainable Financial Health Score, (ii) category-specific transaction anomaly detection, and (iii) progressive expense and cash-flow forecasting. Each output is derived from data the user has supplied, is traceable to an explicit rule or model, and is designed to support the later Digital Twin, planning and advisory features of the system.

II. LITERATURE REVIEW

Uyanaheewa et al. proposed WONGA, a smart personal-finance application that automates expense tracking through bill-image processing, bank-SMS extraction, natural-language processing and calendar-event expense prediction. The system used ANN models for SMS classification, OCR and NER for bill extraction, Multinomial Naive Bayes for expense categorisation, ARIMA for short-term expense forecasting, and DistilBERT with regression models for event-based expense estimation, reporting 97% SMS classification accuracy, 88% transaction-extraction accuracy and 98.93% expense-classification accuracy. Its limitations include dependence on region-specific datasets and the absence of an explainable health score, adaptive income-tier budgeting, anomaly feedback or Digital Twin modelling [1].

Hyndman and Khandakar presented automatic forecasting with exponential smoothing and ARIMA, providing the

statistical basis for FinGear AI's forecasting pipeline. Their work is general-purpose and does not address personal-finance context, transaction classification or explainable recommendations [2].

Anshari, Almunawar and Masri explored Financial Digital Twins for robo-advisory, showing how personal data and analytics can support dynamic, customised advice. The work is conceptual and does not implement a complete transaction-level health-scoring and forecasting system [3].

Bonelli studied Digital Twin personalisation and conversational AI in robo-advisory using a scenario-based randomised experiment with 336 respondents. Digital Twin personalisation improved perceived personalisation but increased privacy concern, while conversational AI improved perceived interaction quality; the study did not evaluate real financial outcomes [4].

Acharya and Bhojak proposed a Financial Doppelgangers framework combining Digital Twins, Graph Neural Networks, Distributed Ledger Technology and Explainable AI for systemic-risk detection in FinTech ecosystems. Evaluated on over 12,000 peer-to-peer lending transactions, it reported 88% early-warning accuracy and AUC 0.91, but targeted ecosystem-level rather than individual personal-finance management [5].

Jiang et al. proposed an LSTM and multi-objective optimisation approach for personalised investment advice using market information, behaviour, goals and risk appetite. Simulated results showed more stable responses to market volatility, but the study did not address real-world suitability or integration with everyday spending behaviour [6].

Kong et al. reviewed Large Language Model applications in finance, covering financial text analysis, sentiment analysis, forecasting, reasoning and agent-based modelling. They highlighted the potential for personalised advice alongside hallucination, reasoning and privacy risks, supporting the use of structured financial data to ground explanations [7].

Das, Markowitz, Scheid and Statman developed a mental-account portfolio framework linking Mean-Variance Theory, Behavioural Portfolio Theory and goal-based risk management. The work demonstrates mathematical treatment of goal-specific portfolios but does not incorporate real-time personal transactions or AI-based behaviour analysis [8].

Leukam, Koffi and Djagba proposed goal-based portfolio optimisation using G-Learning and inverse reinforcement learning, reporting an improvement in simulated Sharpe Ratio from 0.42 to 0.483. The evaluation is simulation-based and does not fully account for real transaction costs, market impact or consumer deployment [9].

Feng, Li and Liu proposed a responsible-AI framework for robo-advisory and financial planning centred on fiduciary

duty, adaptive personalisation, robustness, ethics, fairness and auditability. The work provides a conceptual roadmap rather than a fully implemented financial-planning system [10].

Černevičienė and Kabašinskas reviewed 138 studies on Explainable AI in finance, identifying credit management, stock prediction and fraud detection as common applications and feature importance, SHAP and rule-based methods as frequent explanation techniques. The review also identifies unresolved challenges in explanation quality and fairness [11].

Collectively, the literature provides separate foundations for transaction processing and ARIMA forecasting [1], automatic forecasting [2], Financial Digital Twins and trust [3], [4], [5], goal-oriented investment [6], [8], [9], and Explainable AI [11]. No reviewed system combines adaptive income-tier budgeting, a connected transaction ledger, an explainable health score, per-user category-specific anomaly feedback and progressive forecasting in one architecture; Phase I of FinGear AI addresses this gap.

Collectively, the surveyed literature provides strong but separate foundations: WONGA for transaction extraction, classification and ARIMA-based prediction [1]; Hyndman and Khandakar for automatic forecasting [2]; the Digital Twin studies for dynamic user representations, together with privacy and trust concerns [3]–[5]; the portfolio papers for goal-oriented allocation, generally assuming an already-funded profile [6], [8], [9]; and the Explainable-AI review for interpretable rules in the core health score alongside machine learning for anomaly detection and forecasting [11]. No reviewed system combines adaptive income-tier budgeting, a connected transaction ledger, an explainable health score, per-user category-specific anomaly feedback and progressive forecasting in a single architecture; Phase I of FinGear AI addresses this gap.

III. METHODOLOGY

The Phase I architecture as illustrated in Fig. 1, consists of a React frontend and a FastAPI service layer backed by PostgreSQL, organised into two connected sections: a Financial Analytics System and a Personalized Decision Support layer. Financial profile and transaction data submitted through the frontend pass through JWT-authenticated Application Programming Interface (API) routes, after which a data-processing pipeline performs input validation, transaction categorisation, feature preparation and organisation of user history. PostgreSQL serves as the persistent, user-scoped store for profiles, transactions, health-score history, anomaly feedback and forecast records, ensuring that every analytical engine works from a single, consistent ledger rather than independently maintained copies of the same data.

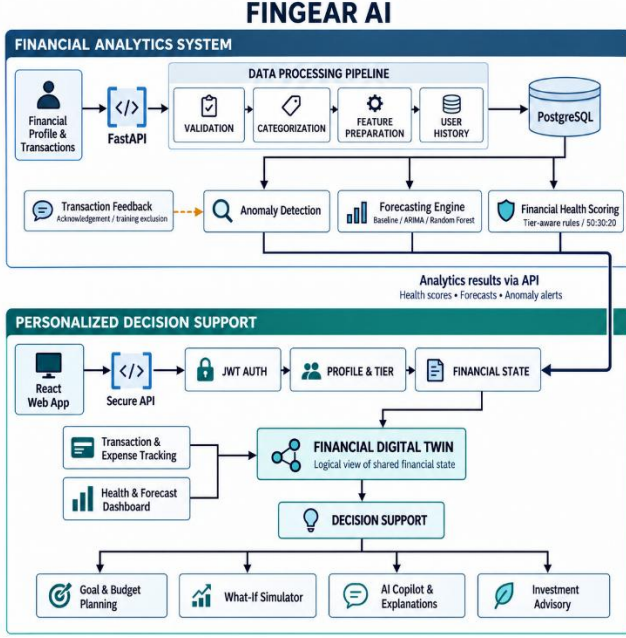


Fig. 1. FinGear AI system architecture (Financial Analytics System and Personalized Decision Support)

Three analytical engines operate on this shared financial context. The anomaly-detection engine evaluates new spending against the user's established category-specific behaviour; the forecasting engine combines statistical baselines, Autoregressive Integrated Moving Average (ARIMA) and a user-specific Random Forest regressor according to the depth of available history; and the financial-health engine applies explainable, tier-aware weighted rules derived from the classic 50:30:20 principle, adapted per income tier, as shown in Table I. Their outputs are returned through the API to the Personalized Decision Support layer, which assembles them into transaction and expense-tracking views and health and forecast dashboards, with goal planning, what-if simulation, AI-Copilot guidance and investment decision support to follow in later phases. The architecture therefore treats the Financial Digital Twin as a logical view over this shared analytical state rather than a separately maintained database.

TABLE I. INCOME TIERS AND TARGET ALLOCATIONS

Income Tier	Monthly Income	N:W:S (%)
Survival	Below ₹30,000	65:15:20
Baseline	₹30,000–₹49,999	50:30:20
Accumulation	₹50,000–₹79,999	45:25:30
Reverse Budget	₹80,000–₹1,24,999	40:20:40
Wealth Building	₹1,25,000 and above	35:15:50

A. Financial Health Score

Needs–wants–savings budgeting [12], the EMI-to-income (debt-to-income) ratio [13] and emergency reserves sized against essential expenses [12], [14] are established personal-finance principles. The contribution of FinGear AI is not new

financial mathematics but their adaptive, explainable integration: targets vary with income tier, and the indicators are combined with safety guardrails into one transaction-driven score.

Users are classified into the five income tiers of Table I, each with a distinct Needs–Wants–Savings (N:W:S) target ranging from 65:15:20 for Survival to 35:15:50 for Wealth Building.

Let I denote average monthly income, NE needs expenditure, WA wants expenditure, SV savings, EMI mandatory debt repayment, OD outstanding debt and EF liquid emergency funds; N_T , W_T and S_T are the tier's target ratios. SS, ER, DM, FR and PD denote Spending Sustainability, Emergency Resilience, Debt Manageability, Future Readiness and Payment Discipline and Protection. The actual allocation ratios are then computed as in (1):

$$N = NE / I, \quad W = WA / I, \quad S = SV / I \quad (1)$$

Spending Sustainability (weight 30%) combines allocation fit and surplus fit. NeedsFit, WantsFit and SavingsFit measure the normalised deviation of the actual allocation ratios from the tier targets, scoring 100 when needs and wants are at or below their targets and savings at or above, as in (2); SurplusFit measures the achieved monthly surplus $MS = I - NE - WA$ relative to the target savings amount $TS = S_T \cdot I$, as in (3). They are combined using (4) and (5):

$$\begin{aligned} NeedsFit &= \min(100, 100 \cdot N_T / N) \\ WantsFit &= \min(100, 100 \cdot W_T / W) \\ SavingsFit &= \min(100, 100 \cdot S / S_T) \end{aligned} \quad (2)$$

$$SurplusFit = \min(100, 100 \cdot \max(MS, 0) / TS) \quad (3)$$

$$AllocationFit = 0.25 \cdot NeedsFit + 0.30 \cdot WantsFit + 0.45 \cdot SavingsFit \quad (4)$$

$$SS = 0.75 \cdot AllocationFit + 0.25 \cdot SurplusFit \quad (5)$$

Emergency Resilience (weight 25%) compares liquid funds against a risk-adjusted target, as defined in (6). EmergencyTarget is the essential monthly expenditure NE multiplied by a reserve period of 3 months for salaried users without dependents, 4 months with dependents, 5 months with more than two dependents and 5 to 6 months for self-employed, freelance or business users, with a minimum floor of ₹25,000 for Survival-tier users:

$$ER = \min(100, 100 \cdot EF / \max(EmergencyTarget, 1)) \quad (6)$$

Debt Manageability (weight 20%) penalises high monthly debt-service pressure and total leverage. With $r_E = EMI / I$ and $r_D = OD / (12 \cdot I)$, the sub-scores are given in (7) and the component in (8):

$$\begin{aligned} EMIScore &= \min(100, \max(0, 100 \cdot (0.40 - r_E) / 0.25)) \\ DebtScore &= \min(100, \max(0, 100 \cdot (1.50 - r_D))) \end{aligned} \quad (7)$$

$$DM = 0.60 \cdot EMIScore + 0.30 \cdot DebtScore + 0.10 \cdot (OnTimeRate \cdot 100) \quad (8)$$

EMIScore is 100 when EMI is at most 15% of income and falls linearly to zero at 40%; DebtScore falls linearly from

100 to zero as outstanding debt rises to 1.5 times annual income.

Future Readiness (weight 15%) evaluates saving-target achievement, goal feasibility and investment consistency, as given in (9):

$$FR = 0.50 \cdot \text{SavingTargetAchievement} + 0.30 \cdot \text{GoalFeasibility} + 0.20 \cdot \text{InvestmentConsistency} \quad (9)$$

SavingTargetAchievement is the achieved savings *SV* as a percentage of the target *TS*, capped at 100. *GoalFeasibility* averages, over active goals, the planned monthly contribution as a percentage of the required one (100 when met, otherwise 10–100). *InvestmentConsistency* is the share of the last three months with an investment contribution. With no active goal, the goal weight is redistributed proportionally.

Payment Discipline and Protection (weight 10%) evaluates on-time payments, planning status and insurance coverage, as given in (10):

$$PD = 0.60 \cdot \text{PaymentScore} + 0.25 \cdot \text{PlanningScore} + 0.15 \cdot \text{ProtectionScore} \quad (10)$$

PaymentScore is the on-time payment rate scaled to 100. *PlanningScore* is 100 when a budget, emergency planning and active goals are all in place, 65 when any two are, and 30 otherwise. *ProtectionScore* is 100 when insurance expenses are recorded and 50 with only provident-fund or fixed-deposit information.

The overall raw score combines all five weighted components as shown in (11), and the guardrails in (12) produce the final Financial Health Score (FHS), which is rounded and restricted to the range 0–100:

$$\text{RawScore} = 0.30 \cdot SS + 0.25 \cdot ER + 0.20 \cdot DM + 0.15 \cdot FR + 0.10 \cdot PD \quad (11)$$

$$FHS = \min(\text{RawScore}, C_s, C_e) \quad (12)$$

Here $C_s = 59$ when $MS < 0$ and 100 otherwise, and $C_e = 49$ when EMI reaches 40% of income and 100 otherwise. These two safety guardrails prevent strong performance in one area from concealing a serious affordability or debt problem. The weights are design parameters rather than statistically derived values: they prioritise spending sustainability and emergency resilience (55% together) while retaining contributions from debt, future planning and payment discipline, and their sensitivity is not evaluated here. Scores are graded Financially Healthy (80–100), Stable/Improving (60–79), Needs Attention (40–59) or Financially Vulnerable (0–39). The engine additionally raises risk flags whenever emergency reserves cover less than one month of essential outflow, a bill or EMI has been missed within the last 90 days, EMI pressure reaches at least 35% of income, or outstanding debt exceeds 1.2 times annual income, giving the user explicit, actionable warnings rather than a single opaque number.

A separate Data Confidence measure, given in (13), is computed independently of the health score itself, so that incomplete records reduce certainty rather than silently lowering the user's result:

$$\text{DataConfidence} = 0.40 \cdot \text{HistoryCompleteness} + 0.40 \cdot \text{CategorisationCompleteness} + 0.20 \cdot \text{IncomeVerification} \quad (13)$$

The system reports High Confidence above 80, Moderate Confidence between 50 and 79, and a Provisional Score below 50 or when history is insufficient. Finally, an adaptive recommendation engine converts the score into practical next steps: excess discretionary spending is redirected gradually rather than all at once, so that the recommended monthly reduction is capped at one third of the excess or 5% of income, whichever is smaller, and any emergency-fund shortfall or required increase in savings is expressed as a concrete monthly target rather than a generic instruction.

B. Transaction Anomaly Detection

Each user maintains an independent anomaly-detection model for every spending category — Food, Transport, Housing, Utilities, Shopping, Entertainment and others — so the same transaction amount may be considered normal in one category and unusual in another for the same person. When a user has insufficient history in a category, a cold-start process based on robust baseline rules and category-level median behaviour is used, allowing anomaly evaluation to begin even with limited data. As more transactions accumulate, the model transitions to a machine-learning ensemble combining category-level clustering (K-Means) with autoencoder-based pattern reconstruction, so that both distance-based and reconstruction-error-based signals contribute to the final anomaly score.

For each evaluated transaction, the module returns the spending category, a normalised anomaly score, a severity level, the current model's maturity status and a short explanation of which behavioural signal triggered the flag. Rather than automatically rejecting a transaction, the module functions as a decision-support layer: the user can confirm that a flagged transaction is normal, in which case it is folded back into the category's learned baseline, or exclude it from future model training, in which case the record remains visible in the ledger for bookkeeping but no longer influences what the model considers typical for that category. This design allows a single unusual but legitimate event — such as a medical bill or emergency repair — to be handled without permanently distorting the model's understanding of the user's everyday spending, and because models are isolated per user and per category, one person's behaviour never affects another user's anomaly baseline.

C. Progressive Expense Forecasting

The forecasting engine selects a model according to the amount and quality of available personal history, moving through three progressive stages rather than committing to a

single model for every user. A statistical/rule-based baseline is used for newly registered users, for categories with fewer than seven distinct expense days, or whenever recent history is incomplete; it produces a cautious estimate from recent average expenses and observed transaction frequency, together with an explicit maturity label so the user understands how much to trust it. Once at least seven distinct expense days are available, a user-specific Random Forest regressor is trained on the individual's own transaction history to capture nonlinear relationships that a simple average cannot represent — for example, higher spending on particular days of the month, weekend spending spikes, category-specific patterns, and interactions between time period and category.

As multi-month time-series data accumulates, a per-user ARIMA model is additionally trained to capture trend and temporal dependency between successive periods, complementing the Random Forest model's ability to capture nonlinear, multi-feature relationships. Before any model is trained, raw transactions are preprocessed into date, type, category, amount, day/week/month, recurrence and recent-trend features drawn from the same validated ledger used by the health-score and anomaly-detection engines, so that all three modules remain consistent with one another. Daily forecasts support short-term planning by estimating the expected expense for upcoming days, while monthly and category-level forecasts aggregate these estimates together with recurring transactions to project cash-flow movement, as shown in (14) and (15):

$$\text{ProjSurplus} = \text{ProjIncome} - \text{ProjNeeds} - \text{ProjWants} - \text{ProjDebtPayments} \quad (14)$$

$$\text{ProjSavingsCapacity} = \max(0, \text{ProjSurplus}) \quad (15)$$

Every forecast is labelled provisional, developing or mature according to the depth of the user's transaction history, the number of distinct expense days, category completeness and the stability of observed spending — so that a prediction built from a few days of data is never presented with the same certainty as one built from an established, multi-month time series. Forecasts are recalculated automatically whenever a new transaction, a recurring-transaction change, an income adjustment, a restored record or a model-training exclusion alters the underlying ledger, so the projected surplus and category-level estimates always reflect the user's current financial condition rather than a stale snapshot.

IV. RESULT AND ANALYSIS

The three Phase I modules were evaluated independently using module-specific benchmarks.

A. Financial Health Score

The scoring engine was tested across all five income tiers, and the implementation achieved 100% alignment with the predefined scoring rules across the evaluated scenarios. Table

II summarises representative results. The critical-debt guardrail also achieved a 100% trigger rate: whenever the tested EMI-to-income ratio reached or exceeded 40%, the score was correctly capped at 49 or below. These results confirm that the implementation follows the specified rules; they do not by themselves establish the financial validity of the scoring framework.

TABLE II. HEALTH SCORE VALIDATION ACROSS INCOME TIERS

Income Tier	Monthly Income	Score	Assessment
Survival	₹22,000	38/100	Vulnerable
Baseline	₹42,000	72/100	Improving
Accumulation	₹72,000	92/100	Healthy
Reverse Budget	₹1,15,000	97/100	Healthy
Wealth Building	₹2,40,000	97/100	Healthy

B. Transaction Anomaly Detection

The ensemble was evaluated on 300 labelled transactions (240 normal, 60 anomalous), achieving 95.67% classification accuracy. All 240 normal transactions were correctly classified and 47 of the 60 anomalies were detected, giving 100% precision, 78.33% recall, an F1-score of 0.8785 and a Receiver Operating Characteristic – Area Under the Curve (ROC-AUC) of 0.9992. No false positives occurred, meaning every flagged transaction in the evaluation was genuinely anomalous, though 13 anomalies remained undetected — indicating a trade-off between alert reliability and recall, as shown by the score distribution and confusion matrix in Fig. 2 and the flagged transactions in Fig. 3.

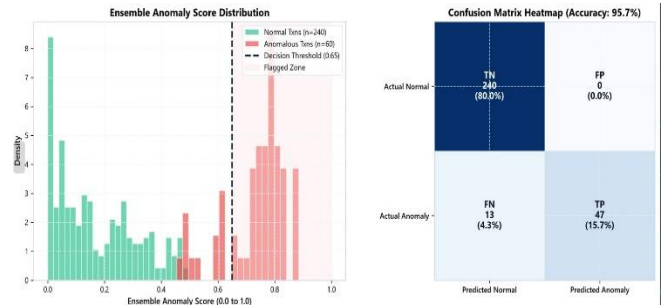


Fig. 2. Anomaly score distribution and classification confusion matrix

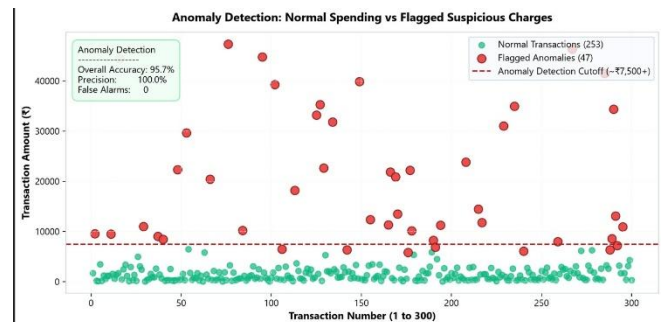


Fig. 3. Transaction spending distribution with flagged anomalies

C. Progressive Expense Forecasting

The Random Forest expense forecaster was evaluated using an 83-day rolling walk-forward back-test, achieving 82.94% forecasting accuracy, a mean absolute error of ₹210.39 per day, a root-mean-squared error of ₹295.88 and an R^2 of 0.5624. Directional accuracy (predicting whether spending would rise or fall) was lower at 48.78%, showing that the model estimates spending magnitude more reliably than short-term direction. Average response time was 44.86 ms under validation conditions, and Fig. 4 compares actual against AI-predicted daily expenses over the evaluated window.

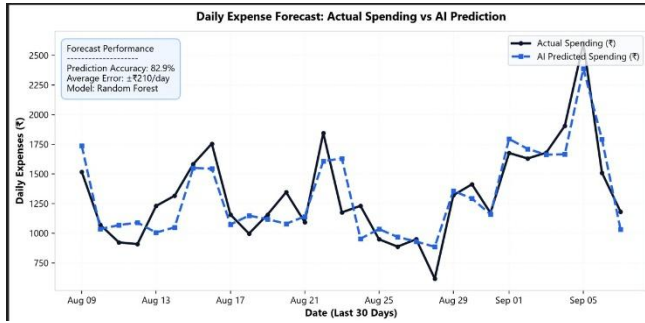


Fig. 4. Actual versus AI-predicted daily expenses over a 30-day window

Overall, the three modules — 100% rule alignment, 95.7% anomaly-detection accuracy and 82.9% forecast accuracy — represent different performance measures and should not be combined into a single score. The scoring engine applied the specified tier rules and debt safeguards consistently across every tested scenario. Anomaly detection produced no false alarms, but its 78.33% recall shows that some unusual transactions were missed, a reliability–coverage trade-off that retraining and threshold tuning could improve. Expense forecasting met its benchmarks for accuracy, average error and R^2 , but its low directional accuracy (48.78%) shows that estimating spending magnitude and anticipating day-to-day movement are distinct capabilities, so forecasts are best read alongside their error measures. Together, the results show that rule-grounded explainability and adaptive machine learning can coexist within a single personal-finance system. The scoring result confirms faithful implementation of the predefined rules rather than the empirical validity of the framework, and the weights remain design parameters whose sensitivity is yet to be studied. Broader validation across additional users, categories and longer time horizons is needed to establish how consistently these results generalise.

V. CONCLUSION

This paper presented the Phase I financial-assessment layer of FinGear AI, comprising an adaptive, explainable Financial Health Score, a per-user category-specific anomaly-detection engine, and a progressive baseline-to-ARIMA-to-Random-Forest forecasting pipeline. Income-tier classification ensures that users with different financial

circumstances are evaluated against suitable targets rather than a single generic rule, while explicit component-level formulae keep every score, alert and forecast traceable to interpretable evidence. Validation results — 100% alignment of the implementation with the predefined scoring rules, 95.67% anomaly-detection accuracy with 100% precision, and 82.94% forecasting accuracy — support the feasibility of this approach. This foundation is designed to support the later Digital Twin features of FinGear AI, including goal and budget planning, what-if simulation, AI-Copilot guidance and investment decision support.

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